Freshly Cooked or Factory-Prepared? Estimating Restaurants' Hidden Production Practices through Labor-Market Signals

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Abstract

Many Chinese consumers worry about the use of prepared meals in restaurants. However, this production practice is usually unobservable. This study identifies a labor-market signal which may indicate the restaurants' reliance on prepared meals. We assemble an original restaurant-level dataset from 5 major cities in China by combining online recruitment postings, commercial-rent listings, chain size, location-based cooking constraints, and consumer reviews. A Multiple Indicators Multiple Causes (MIMIC) model uses the chef-to-server wage ratio and the share of negative reviews mentioning prepared meals as reflective indicators of latent prepared meal reliance. The estimation uses 311 complete restaurant observations. Results show that higher rent, greater chain size, and location in the interior of large commercial complexes are each associated with greater latent reliance. Consistent with the proposed mechanism, reliance is negatively reflected in the chef-to-server wage ratio and positively reflected in prepared-meal-related complaints. The study contributes to consumer-affairs research by offering consumers and regulators a practical means of mitigating information asymmetry and supporting more transparent disclosure of restaurant production practices. The study also points out that opaque production practices in one market may be inferred from observable traces in an adjacent market.

Keywords: Prepared meals; Consumers' right to know; Labor-market signals; Restaurant transparency; MIMIC model; China.

JEL Classification: D82; J31; L83.

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1 Introduction

The Prepared meals industry has grown rapidly in China recent years, especially after the COVID-19 pandemic (Qianzhan Industry Research Institute, 2023; Zhou, 2021). However, the controversy on prepared meals has exploded since 2023 in China. "Wuxi School Prepared Meals Event" in September 2023, where parents suspected that the school had served students meat that had been stored for as long as one year , first raised the concern about prepared meals in China.[1] Then, in 2025, after dining at Xibei, Chinese businessman Luo Yonghao argued that the restaurant used prepared dishes and criticized

[1]For news report, see China News Service, "Did a School Serve Expired Food to Children? Official Response," *Huanqiu*, September 9, 2023, `https://hqtime.huanqiu.com/article/4ET191SCl2N`.

their poor taste.[2] This ignited the long lasting online debate between Jia Guolong, the boss of Xibei, and Luo Yongao. Luo Yonghao then argued for the consumers' right to be informed about the use of prepared meals.[3] After that, Chinese media have constantly reported some restaurants that were found using prepared meals without informing the consumers.[4]

According to industry standard cited by Xue et al. (2025), prepared meals are defined as pre-packaged finished or semi-finished meals made from agricultural, livestock, poultry, and aquatic products through pre-processing or pre-cooking.[5]

Industry estimates suggest that prepared meals had already penetrated approximately 12% of restaurant food procurement in China by 2021, with the penetration rate projected to reach around 20% by 2025 (Tungee Research Institute, 2022). In contrast, the penetration rate in the US was above 60% in 2020, according to Liu (2022, p. 11). Though the penetration rate of prepared food in restaurants is relatively lower than in the US, this has already raised concerns among Chinese consumers.

Concerns about prepared meals originate from various aspects. The first reason is related to the price. Using prepared meals is an effective way to reduce the cost (Zhou, 2021). However, the lack of prior information may cause consumer consumers to pay prices for prepared meals comparable to those charged for freshly cooked meals. The interview conducted by Xue et al. (2025) shows that some Chinese consumers regard prepared meals as "inferior" to freshly cooked meals, and should charge a lower price, but not pretend to be freshly cooked and charge a relatively high price.

Moreover, China has a history of food safety issues, especially concerning food products manufactured through industrialized production processes: Melamine-Tainted Milk Scandal in 2008 [6], Gutter Oil Scandal from 2011 to 2012 [7], Preserved Vegetable and Pork Prepared-Dish Scandal in 2024 [8], and Lead-Contaminated Food Incident in 2025 [9]. These events often involved improper actions by the supervisory authorities, which have led to a lack of social trust. Though prepared meals are not necessarily unhealthy compared with fresh meals, Chinese consumers often associate prepared meals with past food safety scandals (Xue et al., 2025).

Finally, the taste of prepared meals is also important. The Chinese term *Wok Hei* (the breathe of the *wok*) [10], is an indispensable aspect of judging a Chinese dish's taste. Prepared meals are sometimes

[2] See Xue Shasha, Zhu Xuan, Hong Yu, and Wu Qi, "On-site Investigation of Xibei's Preparation of Luo Yonghao's 'Same Menu': The Scallion-Flavored Grilled Fish Used Marinated Frozen Fish with an 18-Month Shelf Life," *The Paper*, September 13, 2025, https://www.thepaper.cn/newsDetail_forward_31602252. See also Shao Bingyan, "Xibei Apologizes: It Will Move as Much Central-Kitchen Pre-Processing as Possible to In-Store Preparation," *The Paper*, September 15, 2025, https://m.thepaper.cn/newsDetail_forward_31612841.

[3] See First Financial, "Xibei's Prepared-Meals Controversy Ignites Industry Debate: Why Are Consumers So Wary of Prepared Meals?", *CCTV.com*, September 13, 2025, https://news.cctv.com/2025/09/13/ARTIXgyJtdBa8iLmImu0ciVt250913.shtml.

[4] For example, *The Beijing News*, "Is the 'Lunch Companion' Going Public? Can RMB 30 Pre-prepared Noodles Retain Office Workers?", October 28, 2025. The report describes Xiao Noodles' standardized production system, in which seasonings, toppings, and noodles are centrally prepared and subsequently assembled at individual restaurants.

[5] Another frequently used term in China is "Central Kitchen Production Mode". Government has defined "central kitchens" as prepared meal factories that are owned by chain restaurants and supply prepared food for themselves since 2024 (State Administration for Market Regulation et al., 2024). This distinction in legal terminology serves as a policy discourse to alleviate consumers' distrust toward prepared meals (Xue et al., 2025). However, since no clear technological boundary can be drawn between prepared meals and central kitchen production, this paper will not do further distinction and will use the term "prepared meals" to refer to both.

[6] Melamine was illegally added to diluted milk to falsify protein-test results. See Reuters, "Factbox: What Is Melamine, and Why Add It to Milk?," January 22, 2009, Reuters.

[7] Waste oil collected from restaurant grease traps, discarded animal products, and other unsanitary sources was illegally processed and resold as edible cooking oil through underground supply chains. See Reuters, "China Arrests 52 over Cooking Oil Scandal," December 20, 2011, Reuters.

[8] Scandal directly related to prepared meals. See CCTV Finance, "Lymph Nodes and Glandular Tissue Clearly Visible: Disturbing Production Conditions behind Packaged *Meicai Kourou*," March 15, 2024, CCTV News (in Chinese; title translated by the author).

[9] A kindergarten allegedly added non-food-grade coloring pigments containing high levels of lead to cakes and other foods. See Xinhua, "Kindergarten Staff Arrested after 247 Children Poisoned with Lead in NW China," July 20, 2025, Xinhua.

[10] The term *wok* refers to a round-bottomed Chinese cooking pan traditionally used for stir-frying, deep-frying, steaming,

considered “lacking of *Wok Hei*”. [11] Chinese consumers regard *Wok Hei* as an important component of the dining-out experience. When restaurants serve prepared meals lacking *Wok Hei*, consumers may feel frustrated (Xue et al., 2025).

It should be noted that our argument does not lie on whether prepared meals are “indeed” inferior to freshly cooked meals. Instead, we recognize that prepared meals may have some advantages in specific dishes (Latoch et al., 2023; Rahman et al., 2023). However, this does not mean that consumers’ call for transparent labeling is unjustified. The legitimacy of consumers’ right to know need not rest on demonstrable differences in product quality. Xue’s (2025) research indicates that Chinese consumers may not resist prepared meals itself but the nontransparent use. Until August 2026, disclosure of the use of prepared meals is not mandatory for restaurants in China.

As mentioned above, using prepared meals can reduce the cost of restaurants. However, since many Chinese consumers resists the use of prepared meals in restaurants, restaurants have incentive to conceal this information. This creates an information asymmetry so that a market failure similar to Akerlof’s (1970) “market for lemons” may occur. Such market failure may impose a “crowd out effect” on freshly cooked meals that are actually valued by consumers (See section 2.1).

Building on the preceeding outline, this study explores signals in labor-market information that can indicate or explain restaurants’ reliance on prepared meals, thereby helping to mitigate information asymmetry and enable informed consumptions.

This study used the chef-to-server wage ratio, $\frac{ChefWage}{ServerWage}$, as an indicator of restaurants’ reliance on prepared meals. When a restaurant applies prepared meals instead of fresh cooking, it essentially outsources some of the cooking tasks to upstream factories or central kitchens (Deng, 2023). We applied Acemoglu and Autor’s (2011) tasks based model to illustrate that such outsourcing can reduce the chef-to-server wage ratio. Since the absolute wages of chefs and servers may be affected by factors such as regional economic conditions and restaurant market positioning, the wage ratio provides a more appropriate estimation on the prepared meals usage than chefs’ wages alone. In practice, consumers have access to wage information by referring to the recruitment platform.

To examine whether labor-market information can reveal restaurants’ hidden reliance on prepared meals, we employ a two-stage empirical strategy that combines a Multiple Indicators Multiple Causes (MIMIC) model with an external validation exercise. The MIMIC model is a structural-equation framework that links observable factors that influence an unobserved latent variable to multiple observable indicators of that latent variable (Jöreskog and Goldberger, 1975; Chen and Jiang, 2019). In this study, the restaurants’ reliance on prepared meals is the latent variable. The MIMIC model is estimated using the full restaurant sample (311 restaurants in five cities) The external validation serves a distinct but complementary purpose. We compare the mean wage ratio of 53 restaurants with publicly documented use of prepared meals, to that of restaurants for which no such disclosure was identified. Taken together, results from MIMIC model and external validation suggest that the wage ratio captures a systematic change in restaurants’ demand for on-site culinary skills and hence serves as a practical, publicly observable indicator of otherwise restaurants’ reliance on prepared meals.

This study makes three major contributions. First, it provides consumers with a low-cost, accessible, and practical signal for assessing restaurants’ hidden reliance on prepared meals. Using publicly available recruitment wages, the chef-to-server wage ratio may help consumers avoid paying freshly cooked meals prices for meals produced through pre-prepared processes. Second, it contributes to research on information asymmetry and market for lemons by showing that opaque production practices in one market may be inferred from observable traces in an adjacent market, as illustrated by the labor-market signal developed in this study. Third, it shifts the perspective of task-based research. Whereas existing studies address how observed offshoring affects relative wages, this study address what observed wage

braising, and other high-heat cooking techniques. *Wok Hei* refers to “the energy a wok bestows upon a stir-fry, giving foods a unique smoky flavor and aroma” (López-Alt, 2012)

[11] Further disscusion on why prepared meals sometimes lack *Wok Hei* can be found in section 3.2.3.

structures can reveal about otherwise unobservable task reallocation, extending the framework to domestic outsourcing and centralized production.

2 Literature Review and Theoretical Background

2.1 Consumer Information Asymmetry

In the catering market, it is difficult for consumers to observe the production mode of the meals. Though we are not able to make such hasty conclusion that fresh meals are better in quality than prepared meals, Chinese consumers do prefer fresh meals to prepared meals and even considered the latter "unhealthy" and "inferior". This gives rise to an information asymmetric problem – only restaurants know the production mode, but have incentive to hide this information.

The market for lemon theory is useful for illustrating the problem of asymmetric information in prepared meals. Akerlof (1970) first provided a canonical analysis of how asymmetric information in the quality of used cars may distort the market. When sellers have more information on the quality of a car, while buyers have to estimate the possible quality by statistics, buyers' valuation of the car based on the expected quality may not satisfy the sellers of good cars, which discourages "good sellers" but encourages "bad sellers". In the long run, this distortion in incentive drives the good products out of the market (Akerlof, 1970).

The similar mechanism may appears in the catering market: since consumers have insufficient information to judge whether the restaurant serves prepared meals before eating, it may be unwilling to pay a full premium for genuinely fresh and skill-intensive cooking. This weakens the market reward for restaurants that persist serving fresh meals with higher costs. In the long run, this distortion in incentive will drive restaurants that persist fresh mode out of the market, which deviates from social preference for freshly cooked meals.

The asymmetric information in the catering market not only originates from nontransparent production, but also from the dishonest advertisement. News reports about restaurants that had advertised or implied freshness but were later reported to rely on prepared, or central-kitchen products have appeared repeatedly in recent years.[12]. Brand T responded to the public challenge of using prepared meals by separating "fresh-live" stores and others. This response is relatively effective in regaining market trust because this separation admitted limited but open use of prepared meals.[13] However, it is difficult to distinguish genuine disclosure from public relations strategies.

Akerlof's discussion of the costs of dishonesty further clarifies that the welfare loss from misleading and dishonest advertisement is not limited to the amount by which buyers are directly deceived. When prepared meals can be presented as fresh meals, dishonest sellers may undermine the viability of honest transactions and drive restaurants who persist fresh mode out of the market (Akerlof, 1970).

Common remedies for lemons-market problems include signaling, screening, warranties, reputation-

[12]For representative media reports, see China News Service's financial outlet Zhongxin Jingwei, "Lengdong rouxian suan bu suan 'xianbao xianzhu'? Yuanji Yunjiao juanru yuzhicai fengbo" [Does frozen meat filling count as "freshly wrapped and freshly cooked"? Yuanji Yunjiao involved in a prepared-food controversy], September 28, 2023, https://www.jwview.com/jingwei/html/m/09-28/560291.shtml; *Beijing News*, "Xibei zhiqian bing xuanbu tiaozheng zhizuo gongyi, 'yuzhicai biaoqian' weihe tieshang nan sixia?" [Xibei apologizes and announces adjustments to its production process: why is the "prepared-food" label difficult to remove?], September 15, 2025, https://m.bjnews.com.cn/detail/1757929395168203.html; Xinhua News Agency, "Zhongxiao xie fabu 2023 nian shida xiaofei weiquan yuqing redian" [China Consumers Association releases the top ten consumer-rights public-opinion issues of 2023], March 15, 2024, https://www.news.cn/politics/20240315/3ee4d07c90c04bae9bc92cc425ef70ea/c.html; and *Workers' Daily*, republished by China News Service, "Gaojia xiyan jin qicheng shi yuzhicai, heyi lingren bushuang?" [Why is it frustrating that nearly 70 percent of an expensive wedding banquet consisted of prepared dishes?], June 7, 2023, https://www.chinanews.com.cn/cj/2023/06-07/10020459.shtml. Accessed July 6, 2026.

[13]See *National Business Daily*, September 16, 2025; *Beijing News*, "Shuaixian shixian zhengcan quan caidan touminghua, 'baogai' hou de Tai Er ruhe chongsu canyin xincheng?" [After its revamp, how does Tai Er rebuild restaurant trust through full-menu transparency?], December 19, 2025; and Tai Er official website, "Brand Story."

building, and third-party certification (Riley, 2001; Dranove and Jin, 2010). Voluntary disclosure of prepared meals usage faces credibility problem. The value of this research includes offering a signal of prepared meals reliance, which is open and accessible for Chinese consumers.

2.2 Consumers' Right to Know

In his Special Message to the Congress, Kennedy's (1962) definition to consumers' informed right not only included protection against fraudulent and misleading information, but also "to be given the facts he needs to make an informed choice". Later research suggested that information in production process, labor condition for example, has effect on consumers' informed choice even without explicitly relating to the quality of the products (Dickson, 2001). Siipi and Uusitalo's (2008) discussion on genetically modified food pointed out that "individuals' food choices are intimately connected to their self-images and world views". The Consumers' Right to Know need not rest solely on demonstrable differences in the physical, functional, or safety attributes of the final product . Information about production processes may have independent value when it enables consumers to make autonomous choices consistent with their deeply held values (Rubel and Streiffer, 2005; Siipi and Uusitalo, 2008).

Discussion in Introduction has already demonstrated the resistance to prepared meals by Chinese consumers. Consumers investigation also indicates that Chinese consumers would make consumption choice based on whether the restaurants use prepared meals. Although Chinese consumers' claims regarding the perceived disadvantages of prepared meals may not always be empirically substantiated, their preferences for traditional cooking methods and culinary craftsmanship nevertheless reflect values that deserve respect. In this regard, requiring restaurants to disclose their use of prepared meals may well be justified.

2.3 The Impact of Production Model Change on Labor Structure

This study draws on the task-based view of production model change to explain why labor-market information may reveal hidden production practices in restaurants.

Acemoglu and Autor (2011) define task as a unit of work activity that produces output, and the output including both service and goods. Different from the canonical skill-based model, treating a job as a fixed bundle of skills, which is insufficiently nuanced to account for the rich relationships among skills, tasks and technologies, the task-based framework distinguishes between workers' skills and the tasks they perform (Autor et al., 2003). Under task-based framework, workers apply their skill endowments to tasks in exchange for wages, and skills applied in tasks produce output (Acemoglu and Autor, 2011). This distinction is important because technological or organizational change may not directly alter workers' skills, but it can change which tasks remain inside the firm, which tasks are simplified, and which tasks are transferred to machines, suppliers, or external production sites. Sampson (2021) based on this task-based framework develops a professional task-automation framework that shows how individual tasks within a given job can be enhanced or disrupted by automation in very different ways, and Kanazawa et al. (2022)successfully applied the task-based model to micro-level productivity analysis through an empirical study of the taxi industry.

Acemoglu and Autor's task-based framework is also especially useful for this study because it explains how production can be reorganized without being immediately visible to consumers. In task-based framework, the labor market are not only simply changed by the variation of productivity, but also changed by the varying allocation of tasks across different skill workers (Acemoglu and Autor, 2011). Acemoglu and Autor (2011) assume that there are three types of skills – low, medium and high – and each worker is endowed with one of these types of skills. In this setting, intermediate tasks are often performed by medium-skilled workers and are more susceptible to automation or relocation than many low-end in-person service tasks and high-end managerial tasks. (Acemoglu and Autor, 2011).

This logic can be applied to restaurant production. A restaurant does not produce meals through a single homogeneous task, but through a series of differentiated tasks, including food preparation, cooking, plating, serving, cleaning (Göde and Ekergil, 2020).

According to Blinder (2007) and Blinder and Krueger (2008), any job that does not need to be done in person (i.e., face-to-face) can ultimately be outsourced, regardless of whether its primary tasks are abstract, routine, or manual. For customer-service-related tasks, although they do not require highly specialized skills, they do require situational adaptability, visual and linguistic recognition, and face-to-face interaction during execution, which makes them non-transferable outside the restaurant setting (Acemoglu and Autor, 2011). For tasks such as cooking or preparation procedures, can be centralized in central kitchens or food processing factories. Once semi-finished or finished food products are produced, they are distributed to restaurants, where the remaining on-site culinary tasks may be reduced to reheating, plating, or final-stage cooking (Deng, 2023).

Therefore, the usage of prepared food or central kitchen production can be interpreted as a form of task reallocation. In a traditional freshly cooked model, a substantial share of intermediate tasks is performed in the restaurant kitchen. Chefs are required to perform cooking-related tasks on site, including preparation, timing, heat control, flavor adjustment, and quality stabilization(China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). These tasks generate demand for culinary labor, especially for chefs with medium skill level who are directly involved in dish production.

However, when a restaurant relies more heavily on prepared food or centralized production, part of culinary tasks are transferred away from the on-site kitchen. The restaurant may still provide similar dishes to consumers, but the task allocation behind those dishes has secretly changed. A chef may retain the same culinary ability, but if fewer valuable cooking tasks remain inside the restaurant, the restaurant has less incentive to pay a high wage premium for that ability. In contrary, the tasks for server will probably remain same after the adoption of prepared meals or central kitchen production mode, so their will likely remain stable, leading to a smaller wage difference between server and chef.

3 Hypothesis Development

It is difficult for consumers or researchers to directly observe restaurants' reliance on prepared meals. However, theoretical discussion on factors that influence the reliance on prepared meals and its indicators is possible and helpful. The following analysis establishes two indicators and three factors, laying a solid foundation for the subsequent empirical analysis.

3.1 Indicators Reflecting Prepared Meals Reliance

3.1.1 Chef-to-server Wage Ratio

The following adaptation is intended to formalize the intuition underlying the proposed wage-ratio indicator rather than validate the reliability of chef-to-server wage-ratio. To formalize this mechanism of the impact of production model change on labor structure, we adapt the task-based offshoring framework in Acemoglu and Autor (2011) to the restaurant context. Let the unit interval $i \in [0, 1]$ represent the full set of production tasks in a restaurant, ordered by increasing complexity. If the i is closer to 0, the tasks are closer to low complexity tasks such as serving the table, plating, reheating. If the i becomes higher, the tasks become more complex, for instance, preparation, timing, heat control, flavor adjustment, and quality stabilization. And finally, when the i is closer to 1, the tasks become the high complexity task in the restaurant, such as managing, developing new cuisines.

As the result, the final product of the restaurants is not a single task, but a combination of a series of tasks.

$$Y_r = \exp\left[\int_0^1 \ln y_r(i)\,\mathrm{d}i\right]$$

In this formula, r represents restaurant, Y_r is the complete dining service that the restaurants provide to customers, and $y(i)$ is the output of task i. And for each single task $y(i)$, the production function is:

$$y(i) = A_L \alpha_L(i) L(i) + A_M \alpha_M(i) M(i) + A_H \alpha_H(i) H(i)$$

In this formula, L is low skill workers, represents the server in the restaurants, M is medium skill workers, represents chef, especially the middle-level chef, in responsible for dishes cooking directly. H is high skill workers, mainly represents the management layer in a restaurant.

A_L $A_M A_H$ represent factor-augmenting technology of different skill workers, while $\alpha_L(i)$ $\alpha_M(i)$ $\alpha_H(i)$ represent productivity of different skill group on $y(i)$, and $L(i)$ $M(i)$ $H(i)$ are number of different skill workers that needs to be assigned to this task.

According to Acemoglu and Autor (2011) Assumption 2, $\frac{\alpha_L(i)}{\alpha_M(i)}$ and $\frac{\alpha_M(i)}{\alpha_H(i)}$ will decrease as i increases, because the more complex the task, the more comparative advantage will be gained from the high-skilled workers compared to the medium-skilled worker, the medium-skilled workers compared to the low-skilled workers.

As the result, there will be two threshold value I_L I_H , which divides the tasks interval into three part: $[0, I_L)$ conducted by low skill workers (servers), (I_L, I_H) conducted by medium skill workers (middle-level chef), $(I_H, 1]$ conducted by high skill workers.

Firms maximize profit by hiring labor until the market wage equals the value of the marginal product of labor (Lumen Learning, 2024). And now, we define $p(i)$ is the price of $y(i)$, and the wage for server, middle-level chef and management layer can be represented as follow:

$$\begin{aligned} w_L &= p(i) A_L \alpha_L(i) \\ w_M &= p(i) A_M \alpha_M(i) \\ w_H &= p(i) A_H \alpha_H(i) \end{aligned}$$

According to the law of one price for skills in Acemoglu and Autor (2011), even though workers of the same skill level are performing different tasks, in equilibrium they will receive the same wage, because if there is a wage difference for different tasks for labor in a same skill group, the firm cannot hire labor for the task with a lower wage in a long term. For all tasks actually performed by the same skill group, the no-arbitrage condition requires them to earn the same wage. Consequently, $p(i)\alpha(i)$ is equalized across the range of tasks assigned to that skill group in equilibrium. So now we define $P_L = p(i)\alpha_L(i)$, $P_M = p(i)\alpha_M(i)$ and $P_H = p(i)\alpha_H(i)$, and rewrite the wage as follow:

$$\begin{aligned} w_L &= A_L P_L \\ w_M &= A_M P_M \\ w_H &= A_H P_H \end{aligned}$$

The Cobb-Douglas technology implies that the "expenditure" in all tasks should be equalized (Acemoglu and Autor, 2011). So we get:

$$p(i)y(i) = Y \qquad \forall i \in [0, 1] \tag{1}$$

The Cobb-Douglas technology let different tasks have the same expenditure, while the wages are equalized in the task interval for the same skill group, the distribution of workers with the same skill level is uniform in their own task interval.

Let us define l, m, h, to be the supply of server, middle-level chef and management layer, so we get $L(i) = l/I_L$, $M(i) = m/(I_H - I_L)$, $H(i) = h/(1 - I_H)$.

As we have already discussed equation (1), the "expenditures" across all tasks are equalized:

$$Y = P_M A_M \frac{m}{I_H - I_L} = P_L A_L \frac{l}{I_L}$$
$$\Rightarrow \frac{P_M}{P_L} = \frac{I_H - I_L}{A_M m} \cdot \frac{A_L l}{I_l}$$

When calculating the wage ratio:

$$\frac{w_M}{w_L} = \frac{A_M P_M}{A_L P_L} = \frac{(I_H - I_L) l}{I_L m} = \left(\frac{I_H - I_L}{m}\right) \bigg/ \left(\frac{I_L}{l}\right)$$

Now consider the usage of prepared meals or the adoption of a central kitchen production model that allows the externalization of a subset of intermediate culinary tasks through centralized kitchens or food processing facilities. Specifically, a task interval $[I', I''] \subset (I_L, I_H)$ can be relocated outside the restaurant. This generates a new task structure in which the effective set of in-house production tasks becomes segmented.

After externalization, the allocation of tasks is given by:

$$T(i) = \begin{cases} L, & i \in [0, I_L), \\ M, & i \in [I_L, I') \cup [I'', I_H), \\ H, & i \in [I_H, 1]. \end{cases}$$

Accordingly, the distribution with the same skill level adjusts to the reduced task:

$$L(i) = \frac{l}{I_L}, \quad M(i) = \frac{m}{(I' - I_L) + (I_H - I'')}, \quad H(i) = \frac{h}{1 - I_H}.$$

This implies that the reallocation of standardized culinary tasks to centralized production facilities reduces the tasks of chefs. As a result, the relative wage structure is affected as follows:

$$WageRatio = \frac{w_M}{w_L} \downarrow \tag{2}$$

In equation (2), w_M represents the wage of middle-level chef and w_L represents the wage of server. This mechanism is consistent with Acemoglu and Autor (2011) implications of offshoring on the structure of wages, providing a task-based theoretical rationale for the predicted direction of why the chef-to-server wage ratio can decline even in the absence of changes in individual skill distributions, as it reflects a contraction in the set of tasks performed by chefs rather than a change in their underlying ability. However, this model just provides the directional prediction used to construct the indicator; the empirical validity of the indicator is subsequently assessed in section 5.

This implication also explains why this study uses the chef-to-server wage ratio rather than the absolute chef wage. The adoption of prepared meals or central-kitchen production mode has two conceptually distinct effects. On the one hand, it externalizes part of the in-house culinary task interval previously performed by middle-skill chefs. On the other hand, it may also operate as a productivity-enhancing technology, since standardized upstream production can reduce unit costs, increase operational efficiency, and raise the effective productivity of restaurant production (China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). Therefore, the direction of the absolute chef wage is theoretically ambiguous. If the productivity effect is sufficiently large, the wage of chefs may remain unchanged or even increase despite the contraction of in-house culinary tasks.

By contrast, the chef-to-server wage ratio is more directly linked to the relative task structure inside the restaurant. When prepared meals relocate a substantial part of cooking, seasoning, heat control, and food-preparation tasks to central kitchens or upstream suppliers, the remaining in-store culinary work becomes closer to simple finishing, reheating, and plating. In the limiting case, the task requirements of in-store kitchen workers become increasingly similar to those of low-skill restaurant workers. According to the law of one price for skills Acemoglu and Autor (2011), workers performing tasks with similar skill requirements should receive similar wages in equilibrium. Hence, as the in-house chef task set contracts and becomes less distinct from low-skill restaurant tasks, the chef-to-server wage ratio should move closer to one.

The wage ratio also reduces the influence of local labor-market and demand-side conditions. Absolute chef wages may vary substantially across cities, neighborhoods, and restaurant segments because of differences in local income levels, consumption capacity, rent, and general wage levels (Combes et al., 2008). A given chef wage level therefore cannot by itself identify whether a restaurant relies heavily on prepared meals. The chef-to-server wage ratio normalizes chef wages by the wage of another occupation within the same local restaurant labor market, making it a more appropriate proxy for changes in the relative demand for in-house culinary skills.

Hence, the relationship of the reliance on prepared meals and the wage ratio is stated as follows:

Hypothesis 1. *Holding other factors constant, the wage ratio of a restaurants is negatively associated with its reliance on prepared meals.*

3.1.2 Comment Ratio

In order to build up the MIMIC model to test the validity of chef-to-server wage ratio, we need to design another indicator, as MIMIC model needs at least two reflective indicators (Jöreskog and Goldberger, 1975). For most consumers, the common way to avoid restaurants that secretly use prepared meals is to look up existing comments of these restaurants. Although no scientific experiments can prove that consumers can accurately determine the difference between prepared meals and freshly cooked meals, some studies have shown that consumers who have already dined in the restaurant before can infer whether the restaurant uses prepared meals secretly by using the degree of processing (Hässig et al., 2023), sensory perception of food (Bai et al., 2019) or unusual serving speed (Xue et al., 2025) as signals.

However, simply focusing on the total number of reviews mentioning prepared meals is not methodologically appropriate. Restaurants with a large number of reviews may naturally receive a greater number of reviews mentioning prepared meals. However, this does not necessarily indicate that they rely more on prepared meals than restaurants that receive fewer such reviews. Therefore, the absolute number of reviews mentioning prepared meals alone may not accurately reflect restaurants' reliance on prepared meals.

Therefore, we define the $CommentRatio$ for restaurant as

$$CommentRatio = \frac{N^{PM}}{N} \tag{3}$$

In equation (3), N^{PM} denotes the number of negative reviews that explicitly mention prepared meals or related production practices (Central Kitchen), and N denotes the total number of negative reviews received by the restaurant. A higher Comment Ratio is consequently expected to reflect greater reliance on prepared meals and should load positively on the latent prepared-meal-reliance construct.

This construction measures the proportion of consumer dissatisfaction that is specifically attributed to prepared-meal production. It has three advantages over a raw review count. First, it normalizes for differences in restaurant popularity and total review exposure. Second, it distinguishes prepared-meal-related dissatisfaction from general dissatisfaction caused by service, price, sanitation, waiting time, or the dining environment. Third, because the numerator is a subset of the negative-review denominator, both are exposed to more comparable review-generation, fraud-detection, complaint, and moderation processes. The ratio may therefore partially absorb restaurant-level differences in the visibility and retention of negative reviews. Moreover, as noted by Jöreskog and Goldberger (1975), in MIMIC model, indicators are conditionally independent given the latent variable. Since $WageRatio$ and $CommentRatio$ arise from substantively distinct information-generating processes: the former reflects labor-market outcomes, whereas the latter reflects consumers' dining experiences. Their joint use therefore provides two independent manifestations through which the proposed latent construct can be evaluated, rather than relying on multiple measures generated by the same underlying information source.

Nevertheless, the indicator is not a direct verification of production practices. Consumer misclassifications and fraudulent reviews may still generate measurement error. We therefore use the $CommentRatio$ only as an auxiliary reflective indicator rather than as a stand-alone classification criterion. In the robustness analysis, we additionally impose minimum-review thresholds to reduce instability arising from small denominators and change $CommentRatio$ into other form to show that the validity of $WageRatio$ does not hinge exclusively on its specific formulation from the consumers' perspective.

Hypothesis 2. *Holding other factors constant, the comment ratio of a restaurant is positively associated with its reliance on prepared meals.*

3.2 Factors that Influence the Reliance on Prepared Meals

3.2.1 Factor 1: Rent-Driven Reliance on Prepared Meals

Unlike many other retail businesses, restaurant operations are commonly divided into front-of-house (FOH) and back-of-house (BOH) areas(O'Neill et al., 2024). The front-of-house area primarily comprises the dining room and other customer-contact spaces, whereas the back-of-house area centers on the kitchen and food-production activities, including storage, preparation, and cooking.

Higher commercial rent increases the opportunity cost of allocating floor space to back-of-house operations. Additional front-of-house space can often be monetized relatively directly by accommodating more tables and customers, whereas back-of-house space contributes to revenue indirectly by supporting food production. This does not imply that kitchen space does not create value. Rather, compared with dining space, kitchen space cannot usually expand customer capacity simply by accommodating additional seats. This spatial trade-off is particularly important in the restaurant industry because customer demand is often concentrated within relatively short meal periods. During peak dining hours, seating capacity and operational throughput therefore become especially valuable.

Traditionally freshly cooked production requires restaurants to maintain sufficient and functionally differentiated back-of-house space for storing ingredients, handling unprocessed raw materials, preparing and cooking food, separating finished products from production waste, and preventing interference or contamination across different production processes (Flessas et al., 2015). By contrast, restaurants that use prepared meals or adopt central-kitchen production can relocate some of these activities to upstream production facilities. Consequently, restaurants that rely on prepared meals can serve the same quantity or even more orders with a smaller kitchen area. Such restaurants may enjoy lower rental costs and increase revenue (Deng, 2023). Industry evidence illustrates the potential magnitude of this cost-saving mechanism. According to Zhou (2021), the rental cost of a typical Chinese dish allocated to each order decreased by approximately 40% when on-site cooking was replaced with prepared meals (see table 1). Therefore, restaurants facing higher rental cost may have a higher incentive to cut the cost by adopting prepared meals.

Table 1: Cost & Profit Comparison of a Typical Chinese Dish

Category	Freshly Cooked	Prepared Meals	Percentage Change
Packaging cost	¥ 2.00	¥ 2.00	
Platform promotional fees	2.40	2.40	
Platform commission	5.20	5.20	
Ingredient cost	6.50	7.50	+ 15.38%
Labor cost	1.60	0.80	− 50.00%
Rental cost	1.00	0.60	− 40.00%
Utilities and miscellaneous expenses	0.40	0.20	− 50.00%
Selling price per serving	20.00	20.00	
Profit per serving	0.90	1.30	+ 44.44%

Note: Percentage change is calculated as (Prepared Meals − Freshly Cooked)/Freshly Cooked.
Source: Authors' compilation based on Zhou (2021).

Accordingly, commercial rent is incorporated into the MIMIC model as a theoretically motivated cause of latent prepared-meal reliance. Hence, the following hypothesis is stated in terms of a systematic association:

Hypothesis 3. *Holding other observed factors constant, a restaurant's latent reliance on prepared meals is positively affected by higher local commercial rent*

3.2.2 Factor 2 (Chain-Driven):

For large chain restaurants, the core challenge is maintaining consistency while expanding geographically (Bradach, 1998). As the result, compared to individual restaurants, large chain restaurants has greater pressure to ensure their consistency and replicability for further expansion. Zhang (2025) states that after introducing central kitchen production mode, firms can improve quality consistency across outlets. According to China Chain Store & Franchise Association and China Renaissance (2022), the trend toward restaurant chain expansion further promotes the application of prepared meals and drives the maturity of the entire supply chain for the central kitchen, and the usage of prepared meals in chain restaurants is already an openly known secret within the catering industry.

Consequently, the central kitchen model meets restaurant chains' cost reduction and efficiency improvement needs (Deng, 2023). Therefore, compared to restaurants with fewer stores or independent restaurants, large restaurant chains appears to have a stronger incentive to use prepared meals.

Accordingly, number of chain stores is incorporated into the MIMIC model as a theoretically motivated cause of latent prepared-meal reliance. The hypothesis is stated in terms of a systematic association:

Hypothesis 4. *Holding other observed factors constant, a restaurant's latent reliance on prepared meals is positively affected by the number of chain stores.*

3.2.3 Factor 3 (Regulation-Driven):

According to the regulations for catering areas in large commercial complexes (Ministry of Emergency Management of the People's Republic of China, 2023, Section 9.2b), restaurants located underground or inside large commercial complexes face stricter fire-safety and facility-layout constraint. Catering establishments shall not use liquefied petroleum gas (LPG) or Class A or Class B liquid fuels. Additionally, kitchen areas using open fire should be arranged against the exterior wall (Ministry of Emergency Management of the People's Republic of China, 2023, Section 9.2e). Facing strict fire compliance pressure, restaurants located in large commercial complexes are more likely to give up open flame cooking.

As mentioned before, Chinese dishes emphasize *Wok Hei*, which to some extent relies on open-fire cooking. Relevant research suggest that the formation of *Wok Hei* is closely associate with the browning enabled by the Maillard reaction (MR) (Ko and Hu, 2020). MR is an important mechanism through which volatile flavor compounds are generated (Liu et al., 2022). MR are highly dependent on surface temperature and moisture conditions. Traditional high-temperature dry-heat cooking methods, such as open-flame cooking, may be more conducive to surface dehydration, temperature increase, and browning (Purlis, 2010). By contrast, studies on microwave suggest that insufficient browning and flavor development are typical limitations of microwave heating (Sumnu, 2001; Dong et al., 2022). These may be the reasons why some Chinese consumers claim that prepared meals that are simply reheated by microwave taste inferior to freshly cooked meals. On the other hand, studies on braised pork show that traditional open-fire cooking can generate a richer profile of characteristic aroma compounds than induction cooking (Wang et al., 2024). Mechanistic accounts of *wok hei* further suggest that open-flame wok cooking is more conducive to smoky and charred sensory notes, as well as aromas associated with lipid oxidation, thermal degradation, and the partial combustion of aerosolized oil (Ko and Hu, 2020; Chin, 2019). Therefore, the good taste of *Wok Hei* needs to be delivered by open-flame cooking, not just the freshness of the ingredients.

Despite open-flame, the packaging, transportation, storage, and reheating of prepared meals may compromise flavor preservation. This is because volatile aroma compounds can be absorbed by packaging materials, while storage and reheating can promote lipid oxidation, protein degradation, and the formation of warmed-over flavor (Sajilata et al., 2007; Chen et al., 2024). Cold-chain temperature fluctuations may further accelerate the loss of desirable volatile compounds and the production of off-flavor compounds (Yu et al., 2020).

Complaints on "gas stove ban" by Chinese chefs occur frequently because they think techniques like "stir-frying" need open flame for rapid and high-intensity heat [14].

Therefore, it is reasonable to make such inference that if gas is banned in large commercial complexes, and if induction cooking cannot provide satisfactory flavor, restaurants may rather use prepared meals to cut the expense on skilled chefs since their skills no longer generate much value for consumer. Incidents in which restaurants adopted prepared dishes and dismissed chefs due to fire-safety regulations have indeed been reported by the Chinese media[15].

Accordingly, whether there is open fire regulation is incorporated into the MIMIC model as a theoretically motivated cause of latent prepared-meal reliance. The hypothesis is stated in terms of a systematic association:

Hypothesis 5. *Holding other observed factors constant, a restaurant's latent reliance on prepared meals is positively affected by open fire ban for restaurants located in large commercial complexes in the interior.*

4 Empirical Method

Our objective is to identify reliable indicators of restaurants' hidden reliance on prepared meals, rather than to estimate the causal effect of prepared-meal adoption on restaurant wage structures. We therefore adopt a latent-variable approach, treating reliance on prepared meals as an unobserved construct that cannot be measured directly. In a MIMIC model, latent constructs are inferred from the systematic

[14] For media reports, see Jenn Harris, "The End of Korean BBQ in L.A.? What the Gas Stove Ban Means for Your Fave Restaurants," *Los Angeles Times*, June 2, 2022, https://www.latimes.com/food/story/2022-06-02/gas-stove-ban-chinese-korean-bbq-electric-new-buildings-restaurants-future; Naoki Nitta, "Would Giving Up Gas Stoves Threaten the Art of Chinese Cooking?," *San Francisco Chronicle*, January 30, 2023, https://www.sfchronicle.com/food/article/gas-stove-wok-chinese-17742332.php; These reports quote Chinese or Asian-restaurant chefs and owners who argue that gas and open flame is important for wok cooking, high-heat stir-frying, and the production of wok hei or smoky flavor.

[15] See Hu Kefei, "The Chef Disappears from the Back Kitchen," *China Newsweek*, January 10, 2022, https://www.inewsweek.cn/viewpoint/2022-01-10/14884.shtml.

relationships among observable variables, even though the constructs themselves remain unobserved (Jöreskog and Goldberger, 1975). The model jointly estimates a set of structural and measurement equations in which the latent variable is linked to both its observed factors and its observable indicators (Frey and Weck-Hanneman, 1984).

In our setting, the latent construct is a restaurant's degree of reliance on prepared meals. It is modeled as being influenced by a set of theoretically motivated factors and reflected in observable indicators. The MIMIC framework is therefore appropriate because it allows us to assess whether the indicator derived from our task-based framework behaves consistently with an underlying construct whose variation is systematically associated with the proposed factors.

The MIMIC model consists of two sets of equation:

$$\mathbf{y} = \lambda\eta + \varepsilon \tag{4}$$

$$\eta = \gamma'\mathbf{x} + \zeta \tag{5}$$

where equation (4) is the measurement model for the latent variable η and equation (5) is the structural equation equation for the latent variable η. In our study, $\mathbf{y}$ is a column vector of 2 indicators of the single latent variable, and $\mathbf{x}$ is a vector of 3 factors of η. In the equation (5), the latent performance is assumed to be caused by the vector of explanatory variable $\mathbf{x}$. ε refers to a vector of zero mean (2×1) measurement error variables associated with the indicators, while ζ is a zero mean scalar structural error that captures unmodeled variables affecting η and the measurement error associated with it. It is assumed that ζ and all the elements of ε are mutually uncorrelated. And the mechanism of the MIMIC model is visualized in figure 1.

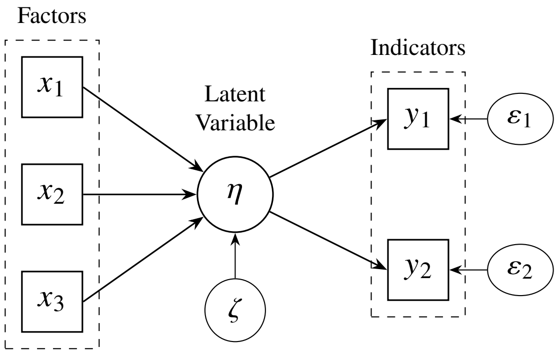


Figure 1: Path Diagram

According to Jöreskog and Goldberger (1975), for a standard scalar MIMIC model, identification requires at least two reflective indicators with uncorrelated measurement errors and at least one observed causal variable. In our MIMIC model, we have 3 factors (observed causal variable) and 2 reflective indicators, which lives up to its identification requirements.

This study assume that a restaurant's reliance on prepared meals is manifested through labor market signal and consumers' experience. We use $WageRatio$ in equation (2) and $CommentRatio$ in equation (3)as indicators of reliance on prepared meals or central kitchen production mode ($\mathbf{y}$) in equation (4).

$$\begin{aligned} WageRatio_i &= \lambda_1\eta_i + \nu_{1i}, \quad \lambda_1 < 0 \\ CommentRatio_i &= \lambda_2\eta_i + \nu_{2i}, \quad \lambda_2 > 0 \end{aligned} \tag{6}$$

We include *Rent*, *Chain* and *Regulation* as the vector of exogenous causal variables (Factors) as $\mathbf{x}$ in equation (5).

$$\eta_i = \alpha_1 Rent_i + \alpha_2 Chain_i + \alpha_3 Regulation_i + \zeta_i \tag{7}$$

where $Rent$ is the daily rent per square meter of restaurant i, $Chain$ is the log of the number of outlets of the brand associated with restaurant i, to account for the potential non-linear relationship between brand scale and its reliance on prepared meal, and $Regulation$ is a dummy variable, if the location of restaurant i makes its usage of open fire become illegal, $Regulation_i = 1$, if not, $Regulation_i = 0$.

All the factor and indicator variables used in our analysis along with their descriptions are summarized in table 2.

Table 2: Definitions of Variables

Variable	**Definition**
Indicators	
WageRatio	Ratio of the wage offered to a mid-level chef to that offered to a server at the same restaurant.
CommentRatio	Number of negative reviews explicitly mentioning prepared meals, central-kitchen production, or related production practices, divided by the total number of negative reviews.
Factors	
Chain	Natural logarithm of the total number of outlets operated by the restaurant chain.
Rent	Daily commercial rent per square meter in the restaurant's local area.
Regulation	Dummy variable, equals to 1 if the use of open flames is prohibited at the restaurant's location, and 0 otherwise.

The aim of using the MIMIC model is to cope with the selection bias of external validation in section 5.2.4, as the external validation based on publicly disclosed central-kitchen adoption is subject to a potential disclosure-selection problem. Restaurants that publicly report the use of central kitchens are unlikely to constitute a random sample of all restaurants using prepared meals. In particular, non-disclosing restaurants may still rely on prepared meals, implying that the comparison group should be interpreted as an unknown group rather than as confirmed non-users.

To address this limitation, the MIMIC model is not built on the publicly disclosed central-kitchen sample. Instead, it uses the full restaurant sample to examine whether the proposed wage-ratio indicator is systematically linked to a latent construct predicted by theoretically motivated economic and regulatory causes. The known-positive central-kitchen sample is therefore used only as an external criterion-validity check, rather than as the primary basis for identifying prepared-meal reliance.

5 Data Analysis and Findings

5.1 Data

This analysis uses data collected in 2026, our original data set contains the wage information of 351 different restaurants in five major cities in China: Guangzhou, Shenzhen, Beijing, Shanghai, and Hangzhou. The initial sample was obtained from BOSS Zhipin, one of China's major online recruitment platforms (Tang, 2016). By using a web-scraping program, we collected restaurant job postings in which the same restaurant simultaneously recruited both front-of-house servers and back-of-house kitchen workers. After removing duplicated postings, non-restaurant businesses, the final labor-market sample contains 351 restaurants.

For each restaurant, we recorded the advertised server wage and the advertised kitchen-worker wage. According to Audoly et al. (2024), for job advertisements providing pay-related information, most of them explicitly revealing the actual salary information. Unlike countries with established tipping systems such as the United States, restaurant servers in China generally do not receive gratuities as a systematic component of compensation (Liu, 2003). Therefore, posted wages provide a closer approximation of their labor-market returns.

Since kitchen jobs differ substantially in skill content, we manually classified chef-related positions into three categories based on job titles and stated job requirements. Junior kitchen positions include tasks such as cutting, dish washing, and other auxiliary back-kitchen work. Middle-level chef positions refer to jobs directly responsible for cooking and dish preparation. Senior chef positions include managerial or administrative responsibilities, such as executive chef, and kitchen-management roles. The empirical analysis uses the wage of middle-level chefs to construct the wage ratio, because this category is most directly exposed to substitution by prepared meals or central-kitchen production, similar to Acemoglu and Autor (2011) definition of medium skill workers. Junior kitchen workers do not fully represent skilled cooking labor, while senior chefs often combine culinary work with management functions. Middle-level chef wages therefore provide the cleanest measure of the labor-market value of restaurant-level cooking skills. Since most job postings give an interval of wage rather than a specific wage, we adopt the midpoint of the interval. This procedure was applied consistently to both server and chef positions.

Following the approach of Boeing and Waddell (2017), who utilized web-scraped online rental listings to construct fine-grained spatial measures of local market conditions, we collect commercial rental information from Anjuke, one of the largest online real estate platforms in China that provides extensive listings of commercial properties. A web-scraping program was used to obtain rental prices and geographic coordinates of commercial spaces. These rental observations were then geographically matched with restaurants identified from BOSS Zhipin to construct a restaurant-level measure of rental cost pressure.

Following prior research that measures restaurant chain scale using publicly available outlet information (e.g., company disclosures and franchise records), we constructed a chain-size measure based on the number of outlets operated by each restaurant brand (China Chain Store & Franchise Association and China Renaissance, 2022; Zhang, 2025). Since outlet information is not uniformly disclosed across Chinese restaurant brands, we adopted a hierarchical manual search procedure. Specifically, we first collected outlet numbers from official brand websites. When such information was unavailable, we relied on Jiamengxing, a Chinese franchise-information platform. Finally, for brands lacking both sources, we manually retrieved store counts from Dianping.

In addition to labor-market and operating-environment data, we manually collected consumer-review information and restaurant consumption per capita from Dianping, one of the largest Chinese online restaurant review platforms, providing user-generated reviews and restaurant information that have been widely used in empirical studies of restaurant markets (Wu et al., 2015). As Dianping explicitly restricts automated scraping, this part of the dataset was collected manually. For each restaurant labor-market sample, we recorded per-capita consumption, the total number of reviews, the total number of negative reviews, and the number of negative reviews that explicitly criticized prepared meals, semi-finished dishes, reheated food, central-kitchen production, or similar standardized food-preparation practices. In total, we collected 24,615 negative reviews, among which 1,339 were identified as involving prepared-meal-related criticism. To reduce measurement error, Dianping reviews were independently collected by three researchers and reevaluated if detected any difference.

After excluding restaurants with missing or unreliable information on rental costs, customer reviews, outlet numbers, or consumption levels, we obtained a final dataset comprising 311 restaurants. This sample size is considered adequate for estimating the parsimonious MIMIC model employed in this study. With two reflective indicators and three observed causes, the reduced-form coefficient matrix is a 2×3 matrix constrained to have rank one, generating $(2 - 1) \times (3 - 1)$ testable restrictions and rendering the model over-identified (Jöreskog and Goldberger, 1975). Given the five observed variables and the two degrees of freedom, a full mean-and-covariance formulation implies approximately 18 freely estimated parameters, corresponding to an observation-to-parameter ratio of $17.3 : 1$. This ratio is substantially above the conventional minimum of approximately five observations per estimated parameter, while the total sample also exceeds the general benchmark of 150–200 observations recommended for structural equation models (Jöreskog and Goldberger, 1975). Moreover, previous applied MIMIC studies have estimated models with 2 or 3 indicators using samples of 288 (Lafuente-Ruiz-de Sabando et al., 2019) and 200 (Barrales-Molina et al., 2013). Therefore, a sample size of more than 300 observations is considered adequate for the estimation of the model.

For the external validation, we identified 21 restaurant brands with publicly documented use of prepared meals or central-kitchen production, based on either corporate disclosures or authoritative media reports. Of the 351 restaurants in the original labor-market dataset, five outlets belonged to these brands and were classified as known positives, leaving 346 restaurants in the no-public-disclosure group. We then supplemented the known-positive sample with wage information from 48 additional outlets of the identified brands. The final validation sample therefore consisted of 53 known-positive and 346 no-public-disclosure restaurants, for a total of 399 unique observations.

5.2 Findings

This Subsection is divided into four parts: section 5.2.1 reports the descriptive statistics, section 5.2.2 reports the results for the MIMIC model, and the robustness results are reported in section 5.2.3. Section 5.2.4 uses data from restaurants which have already voluntarily disclosed their use of prepared meals or have been disclosed by authoritative medias to conduct external validation.

5.2.1 Descriptive statistics

Table 3: Descriptive Statistics

Variable	**Mean**	**Median**	**Standard Deviation**
Chef Wage	7.079	7.000	1.500
Server Wage	5.423	5.500	0.921
Indicators			
Wage Ratio	1.335	1.273	0.320
Comment Ratio	0.082	0.039	0.122
Factors			
Chain	2.277	1.386	2.338
Rent	6.417	4.670	5.534
Regulation	0.277	–	–

Note: Wage variables are measured in thousand RMB per month, but Rent is measured in RMB/m^2/day

Table 3 indicates the summary statistics of the wages, factors, and indicators used in the analysis. Comparing the wages of servers and chefs, the sample mean wage of chefs appears higher than that of servers, which is consistent with the traditional division of labor in restaurants where cooking tasks generally require more specialized skills. Meanwhile, the standard deviation of chef wages is substantially larger than that of server wages.

This pattern is consistent with our theoretical expectation that the adoption of prepared meals may affect different restaurant tasks asymmetrically. While service tasks remain highly dependent on in-person interaction and are relatively difficult to substitute, some cooking tasks, particularly standardized and repetitive preparation tasks, can potentially be replaced by prepared meals or centralized production. As a result, restaurants relying more heavily on prepared meals may exhibit greater heterogeneity in their demand for cooking skills, leading to wider dispersion in chef wages.

In addition, the descriptive statistics of the factors provide preliminary evidence on the heterogeneity of restaurants in terms of operational scale, local cost pressure, and regulatory constraints. These variables are subsequently incorporated into the MIMIC model to examine the factors associated with restaurants' reliance on prepared meals.

5.2.2 Results from the MIMIC model

We present the estimates from the MIMIC model in table 4. Results from three models are presented. The results indicate that all the factors have expected signs and significance. As shown in table 4, the coefficients of rent, chain scale, and regulation are all positive and statistically significant at 1 percent level, indicating that higher rents, greater chain scale, and open-fire cooking restrictions are positively associated with restaurants' reliance on prepared meals.

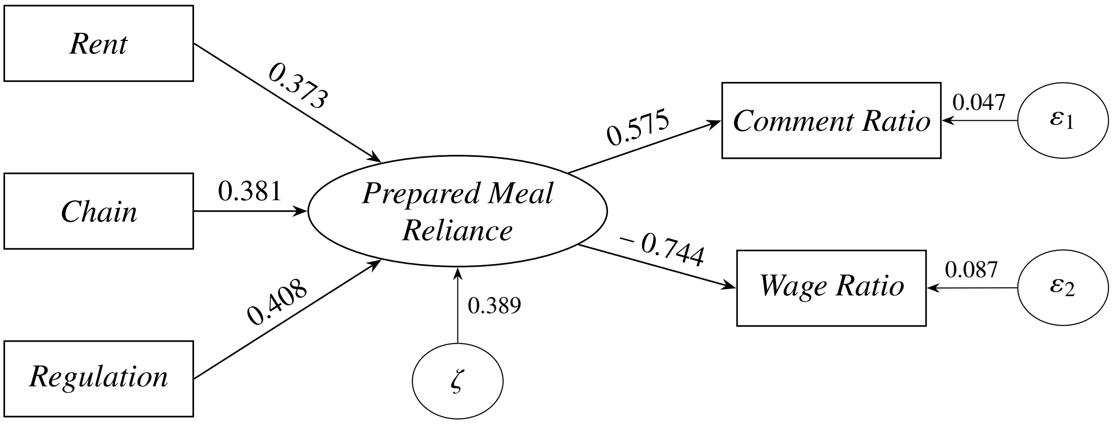
Table 4: Results from MIMIC Model

Variable	**Unstandardized Coefficient**	**Standard Error**	P	**Standardized Coefficient**	**95% CI**
Indicators					
Comment Ratio	1(constrained)	-	-	0.575	-
Wage Ratio	-3.408^{***}	0.419	0.000	-0.744	[−4.229, −2.586]
Factors					
Rent	0.004^{***}	0.000	0.000	0.373	[0.003, 0.006]
Chain	0.011^{***}	0.002	0.000	0.381	[0.007, 0.015]
Regulation	0.063^{***}	0.011	0.000	0.408	[0.041, 0.086]
Diagnostic statistics	**Value**				
Observations	311				
χ^2 [p-value]	3.307 [0.191]				
CD	0.611				
CFI	0.994				
TLI	0.978				
RMSEA	0.046				
SRMR	0.018				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

Diagnostic statistics of the estimated MIMIC model is reported in table 4. A necessary condition for model identification is: $(p \times q + 1/2(p)(p+1) - 2p - q) \geq 0$ where p denotes the number of indicator variables and q is the number of causal variables. According to Jöreskog and Goldberger (1975), the likelihood-ratio test jointly examines whether the reduced-form coefficient matrix satisfies the rank-one restriction implied by the MIMIC model and whether the residual covariance matrix is consistent with a common latent factor and indicator-specific measurement errors. Under the null hypothesis this test statistic is distributed as a χ^2 with $q(p-1) + \frac{1}{2}p(p-3)$ degrees of freedom. In our model, $p = 2$ and $q = 3$, yielding two degrees of freedom. The rank-one restriction is substantively important for the present study because it requires the effects of the observed factors on the two indicators to be proportional to a common loading vector (Jöreskog and Goldberger, 1975). In other words, $Rent$, $Chain$, and $Regulation$ cannot generate arbitrary and unrelated patterns of association with $WageRatio$ and $CommentRatio$ while remaining consistent with the specified MIMIC structure. Indicator-specific effects of observed causes represent additional direct paths outside the standard latent-factor channel and can be explicitly modeled in an extended MIMIC specification. Indicator-specific pathways that affect only one indicator, or affect the two indicators in substantially different proportions, would generally violate the common-factor structure and contribute to model misfit. Thus, the MIMIC model does not require every conceivable direct pathway to be ruled out individually; rather, it tests whether the joint relationships among the multiple causes and indicators are consistent with the restrictions implied by a common latent construct.

The second row of diagnostic statistics in table 4 reports the result from the over-identification test. Rejection of the null hypothesis would suggest the estimated model is mis-specified. $\chi^2 = 3.307$ with its $p = 0.191$, as stated row 2 of table 4, clearly indicate that the estimated model is not mis-specified. Given this evidence, we then provide a series of model fit diagnostics to render the estimated model statically adequate. We use five different statistics: the coefficient of determination (CD), the comparative fit index (CFI), the Trucker-Lewis index (TLI), the root mean square of approximation (RMSEA) and the standardized root mean square of residual (SRMR). These statistics indicate a measure of overall good-fit of our estimated model.



Note: Standardized coefficients are reported. The values associated with the disturbance terms represent standardized residual variances.

Figure 2: Standardized Solution of the MIMIC Model

5.2.3 Robustness check

Since the reliability of indicator $CommentRatio$ may be compromised for restaurants with only a small number of reviews, we exclude restaurants with fewer than 100 total reviews and re-estimate the model using the restricted sample. The results of this model is depicted by table 6. The results remain consistent.

However, the reliability of $CommentRatio$ may be affected by its dependence on the total number of negative reviews. For example, a restaurant with 20 prepared-meal-related reviews out of 40 negative reviews and a restaurant with only 1 such review out of 2 negative reviews would both have a $CommentRatio$ of 0.5. Nevertheless, the former provides substantially stronger evidence of reliance on prepared meals because the indicator is supported by a much larger number of relevant observations. To address this limitation, we replace $CommentRatio$ in equation (6) with N^{PM} which denotes the absolute number of negative reviews that explicitly mention prepared meals, central-kitchen production, or related production practices. The result of this model is depicted in table 7.

The alternative specification remains excellent model fit, and the signs and statistical significance of the measurement and structural coefficients remained consistent with those of the baseline model.

To address potential heterogeneity across restaurant formats, we conduct an additional robustness check by excluding fast-food and hotpot restaurants. These two categories may exhibit inherently different production structures: fast-food restaurants often rely on standardized production systems, while hotpot restaurants require relatively limited cooking labor (China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). Therefore, including them may mechanically reduce the $WageRatio$ independent of prepared-meal reliance. After exclusion, the estimated relationships remain stable. According to table 8, the model remains acceptable after excluding fast-food and hotpot restaurants, with the likelihood-ratio test failing to reject the model specification. All the factors remain positively associated with latent reliance on prepared meals, while $WageRatio$ remains a significant negative indicator loading. Moreover, most of the diagnostics statistics continue to perform well. Although RMSEA increases to 0.100, this measure can be unstable in models with low degrees of freedom and small sample (Jöreskog and Goldberger, 1975).

One potential concern is that the wage ratio may simply reflect differences in restaurants' market positioning or price levels. A potential alternative explanation is that high-end restaurants pay relatively higher wages to front-of-house servers because they require more intensive service, more formal dining procedures, and better customer interaction. Under this interpretation, a lower chef-to-server wage ratio may simply reflect higher server wages in expensive restaurants, rather than a reduction in skilled back-kitchen labor caused by prepared-meal adoption.

To address this concern, we use restaurant-level consumption per capita as a placebo outcome. If the wage ratio primarily reflects restaurant quality, luxury positioning, or high-end service intensity, it should be systematically associated with consumption per capita. In particular, if expensive restaurants have relatively higher server wages, then the chef-to-server wage ratio should be negatively associated with consumption per capita. By contrast, if the wage ratio captures back-kitchen deskilling associated with prepared-meal reliance, rather than general restaurant price level, its relationship with consumption per capita should be weak or statistically insignificant.

Formally, we estimate the following placebo outcome regression:

$$ConsumptionPerCapita_i = \theta_0 + \theta_1 WageRatio_i + \varepsilon_i,$$

where $ConsumptionPerCapita_i$ denotes the average spending per customer at restaurant i.

As reported in table 9, the estimated coefficient of Wage Ratio is statistically insignificant ($p = 0.592$), indicating that the plausibility that the indicator simply reflects restaurant tier rather than differences in production practices is reduced.

5.2.4 External Validation of the Wage-Ratio Indicator

Although the MIMIC model provides internal construct-validity evidence that the $WageRatio$ behaves consistently with the latent reliance on prepared meals, the interpretation of this indicator can be further strengthened through external validation. The purpose of this section is not to prove a causal effect of central-kitchen adoption on restaurant wage structures. Rather, it examines whether restaurants that actively or passively admit their use of prepared meals or central kitchen exhibit the theoretically expected wage-structure pattern.

We construct a validation group based on restaurants' official website or authoritative media's report. A restaurant is classified into the known-positive group if official websites, franchise documents, prospectuses, annual reports, media interviews, or other public materials explicitly disclose that the restaurant adopts a central-kitchen production system, uses standardized semi-finished products, or relies on centrally prepared meal components.

Let C denote the set of restaurants that publicly disclose the use of central-kitchen or prepared-meal production. We define:

$$CentralKitchen_i = \begin{cases} 1, & i \in C \\ 0, & i \notin C \end{cases}$$

It should be emphasized that $CentralKitchen_i = 0$ does not imply that restaurant i does not use prepared meals or centralized production. It only indicates that no public disclosure of the use of prepared meals is available. Therefore, the validation sample should be interpreted conservatively: restaurants with $CentralKitchen_i = 1$ serve as a known-positive group, while the remaining restaurants are not treated as confirmed non-users.

As shown in table 5, group C has a significantly lower mean $WageRatio$ than the no-public-disclosure group. The difference is both statistically significant ($p < 0.001$) and large in magnitude (Cohen's $d = -1.097$), providing external support for the validity of the wage-ratio indicator.

Table 5: External Validation of the Wage-Ratio Indicator

Group	**Observations**	**Mean**	**Standard Deviation**	**Standard Error**
Known-positive group ($CentralKitchen = 1$)	53	1.027	0.069	0.009
No-public-disclosure group ($CentralKitchen = 0$)	346	1.347	0.311	0.017

Between-group comparison	**Value**
Mean difference	-0.319
Welch's t	-16.635
Degrees of freedom	358.912
p-value	< 0.001
95% CI	$[-0.357, -0.282]$
Cohen's d	-1.097
95% CI for Cohen's d	$[-1.395, -0.797]$

Note: The original dataset consists of 351 restaurants. Among them, five restaurants were identified as the Known-positive group based on publicly disclosed information regarding central kitchen or prepared-meal usage. The remaining 346 restaurants constitute the no-public-disclosure group. Mean differences are calculated as $CentralKitchen = 1$ minus $CentralKitchen = 0$. Welch's t-test is reported because Levene's test rejects the equal-variance assumption (F=48.202,p¡0.001).

To further address the concern that the external validation results may be driven by multiple outlets from the same brand, we conduct an additional robustness check using a brand-level de-duplicated sample. Specifically, for brands with multiple publicly disclosed outlets, only one outlet is retained to ensure that each brand contributes only one observation. The resulting sample contains 21 known-positive restaurants without brand duplication.

As shown in table 10, the results remain consistent, this confirms that the observed difference is not driven by the over-representation of particular brands in the validation sample.

6 Discussion and Conclusions

6.1 Discussion

Findings from the analysis indicate that H1-H5 were to supported by the data. This study shows that the wage ratio is negatively correlated with a restaurant's reliance on prepared meals. In addition, the comment ratio reflecting the consumers' complaint on prepared meals, is found to be positively associated with prepared meals reliance, which is consistent with our intuition. Moreover, this study confirms that the chain size, rent, and strict fire regulation can affect a restaurant's prepared meals reliance in positive direction.

Data analysis confirms the nagetive relationship between restaurants' reliance on prepared meals and wage ratio. The reason lies in the outsourcing of tasks of chefs. In section 3.1.1, we applied Acemoglu and Autor's (2011) task based model to illustrate that the chef-to-server wage ratio will decrease when restaurants applies prepared meals to outsource some tasks of chefs to upstream factories. This core hypothesis is the foundation of using wage ratio as an indicator of restaurants' reliance on prepared meals.

Comment ratio is positively related to prepared meals reliance. Recall that the comment ratio is defined as the proportion of consumer comments that mentioned the "prepared meal" in total negative comments on the online review platform. Therefore, it is intuitive that the comment ratio is positively related to a restaurant's reliance on prepared meals. Accordingly, in the MIMIC model, the loading of comment ratio is normalized to 1 to establish the scale of the latent construct. While the normalized loading itself is not statistically estimated, the validity of this hypothesis is supported by the remaining model estimates, whose directions and significance are consistent with our theoretical predictions.

A higher commercial rent may push the restaurant to apply prepared meals. Using prepared meals can effectively reduce costs and the necessary kitchen area. Therefore, under higher rent pressure, restaurants may be more willing to apply prepared meals. Moreover, The chain size has positive effect on the use of prepared meals. This may due to the demand for unifying flavor across different stores of a brand. Finally, the data analysis has confirmed that strict fire-safety regulation in large commercial complexes may lead to higher reliance to prepared meals.

Despite the MIMIC model, the external validation, which is a comparison between the wage ratio of those restaurants publicly disclosed the use of prepared meals, and wage ratios of others in the complete sample. The results indicates that restaurants publicly disclosed the use of prepared meals have a wage ratio significantly lower than others. The external validation also supports the main argument of this study that wage ratio may serve as a indicator of restaurants' reliance on prepared meals.

6.2 Contribution to Practice

Typical consumers have very limited access to the kitchen and have little information on whether restaurants use prepared meals. As mentioned in Introduction, this information asymmetry may cause a similar market distortion as the "market for lemon": Most consumers give high valuation to fresh meals, but lack of information stops them from paying sufficient price for restaurants genuinely persisting fresh cooking, which drives these restaurants out of the market.

This study mainly contributes to consumers' right in practice by mitigating the impact of information asymmetry:

Firstly, in terms of consumers' choice, the method proposed in this study may be a practical guide for consumers. As with other indicators or signals, measuring restaurant reliance on prepared meals is ambiguous and unreliable. Restaurants do not have the incentive to actively disclose their reliance on prepared meals. The chef-to-server wage ratio may help consumers identify restaurants that may highly rely on prepared meals . Though a restaurant's reliance on prepared meals cannot be simply determined by wage ratio, we believe that consumers can refer to wage ratio as an estimation if they do mind having prepared meals in restaurants.

Secondly, with respect to possible future disclosure policy on the use of prepared meals implemented by the government, regulators can refer to the wage ratio to develop a rough estimation and conduct more targeted supervision of restaurants suspected of concealing their use of prepared meals. In addition, responsible government agencies with taxation records may use more precise wage data to regulate restaurants, in comparison to recruitment wage. For those catering enterprises that have not actively declared the use of prepared meals but whose wage ratios approach 1, precise targeted spot checks can be implemented.

It should be noted that the present analysis does not establish a universal numerical threshold above which a restaurant can be conclusively classified as using prepared meals. Accordingly, the indicator should be interpreted as a screening tool rather than definitive proof, and any determination of deliberate consumer deception would require additional corroborating evidence, for example, evidence on supply chain that can confirm the purchase of prepared meals.

6.3 Contributions to Research

The first contribution to research lies in the information asymmetry. By proposing a new indicator to restaurants' hidden production practices, this study shows that hidden information in production practice may be revealed by labor structure. Existing studies on information asymmetry in catering industry mainly focused on voluntary disclosure, mandatory disclosure, or government supervision. This study offers a new perspective on information asymmetry by showing that, even when production practices are not directly observable or are deliberately concealed, the production process may still leave observable traces in adjacent markets — in this case, the labor market. By examining these market signals, consumers and regulators may infer otherwise hidden production arrangements and make better-informed assessments of product value. In this way, labor-market information can help mitigate the information asymmetry problem underlying the market for lemons.

Secondly, this study extends the task-based framework proposed by Acemoglu and Autor (2011). Most applications of this model stressed on explaining or predicting the change in relative wage or inequity of income under the impact of technological and offshoring impact. This study reversely uses the relative wage change to identify change in the production organization, when the latter is unobservable or concealed deliberately. This extends the application of task-based model and can be generalized to other sectors.

Thirdly, this study contributes to the consumer affairs literature by broadening the empirical source of consumer information. Previous research on consumer information focused mainly on mandatory or voluntary product disclosures, firm-generated communications, and third-party information provided by experts or certification systems, and other intermediaries. Although labor-related information has also been examined, it has generally been approached through the lens of consumers' concerns about workers' rights, welfare, and fair treatment[16].

This study contributes to the literature by linking labor-market information to the attributes and production processes of the products consumers purchase. Rather than treating labor information solely as an indicator of worker welfare, we demonstrate that it can also serve as a diagnostic signal of otherwise unobservable production practices that directly affect consumers' evaluations of the product itself. In doing so, this study broadens the empirical sources and inferential mechanisms of consumer information and offers a new avenue for future research in consumer affairs.

6.4 Limitations and Suggestions matter for Future Research

To our knowledge, this is the first study to develop a quantitative measure of restaurants' reliance on prepared meals. The chef-to-server wage ratio introduced in this study provides future researchers with a potentially useful micro-level identification measure for estimating the penetration of prepared meals in China's catering industry.

However, a potential limitation of this study is that, when the method of referring to wage ratio becomes widespread among consumers, even though catering enterprises cannot conceal this information, but they can take some measures to make the wage information more ambiguous. Although directly modifying the recruitment wage may disturb the recruitment, we cannot completely exclude the possibility. Restaurants may set a wide range of advertising wages for both chefs and servers in order to prevent consumers from getting useful information.

It also should be emphasized that this study does not claim that restaurants with a wage ratio below a particular threshold necessarily use prepared meals. A restaurant's reliance on prepared meals cannot be determined solely from its wage ratio, because the ratio may also be affected by other factors. Instead, the wage ratio should be interpreted as a screening signal available to consumers and regulators. Consumers may use it as supplementary information when making dining decisions, depending on their own acceptance of prepared meals.

[16] Consumer evaluations may extend beyond the private benefits to the treatment of the workers involved in its production. Dickson (2001), for example, examined whether apparel consumers would use a "No Sweat" label certifying specified working conditions during garment production and identified a consumer segment for whom such information substantially influenced purchase decisions. Similarly, De Pelsmacker et al. (2005) examined consumers' willingness to pay for fair-trade coffee, treating the fair-trade label as an ethical product attribute that signaled more favorable compensation for coffee producers.

Due to data availability constraints, we rely on cross-sectional data collected within a single period rather than panel data tracking restaurants over time. Although the cross-sectional design allows us to identify systematic differences across restaurants, it cannot fully capture changes in restaurants' production practices or establish dynamic relationships between operational factors and reliance on prepared meals. Future research could construct panel datasets by continuously tracking restaurants' labor-market signals, review information, and operational characteristics over multiple periods. Such longitudinal data would enable researchers to identify appropriate classification thresholds, evaluate the sensitivity and specificity of the proposed indicator, and examine how restaurants' reliance on prepared meals evolves over time, thereby enhancing its practical applicability.

6.5 Conclusion

This study addressed Chinese consumers' concerns about prepared meals. Currently, consumers have limited access to the information about restaurants that using prepared meals before eating in a restaurant. This study finds a systematic negative relationship between the use of prepared meals and the chef-to-server wage ratio. Therefore, wage ratio of a restaurants may serve as an indicator to a restaurant's reliance on prepared meals.

Since the reliance of prepared meals of a restaurant is an unobservable latent variable, this study applies the MIMIC model as the main empirical strategy. We inferred from economic mechanisms that rent, chain size and strict fire-safety regulation may positively affect the restaurants' willingness to use prepared meals, which may then be reflected by comment ratio and wage ratio. Linking factors and indicators of prepared meals reliance through the MIMIC model, we are able to test the hypotheses. The findings from data support our hypotheses in both significance and direction of the estimated effect. This study also used wage data from restaurants that publicly disclosed their use of prepared meals for external validation. The result also indicates that restaurants which use prepared meals have significantly lower wage ratio. Both MIMIC model and external validation verify that wage ratio does negatively correlate with the reliance on prepared meals.

This study supports consumer interests by providing consumers and regulators (if future disclosure policy is to be taken) with a possible indicator to make rough estimation on restaurants' reliance on prepared meals. Future research could further refine the wage-ratio indicator by establishing a clearer quantitative relationship between its values and the degree of reliance on prepared meals. Moreover, by applying Acemoglu and Autor's (2011) task based model to illustrate the economic mechanism, this study indicates that the hidden production practices can be inferred from unhidden labor market information, thus can be an important source of consumer information.

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A Tables

Table 6: Results from MIMIC model excluding all the restaurants with reviews less than 100

Variable	**Unstandardized Coefficient**	**Standard Error**	*P*	**Standardized Coefficient**	**95% CI**
Indicators					
Comment Ratio	1(constrained)	-	-	0.615	-
Wage Ratio	-3.750^{***}	0.505	0.000	-0.730	$[-4.741,\ -2.760]$
Factors					
Rent	0.004^{***}	0.000	0.000	0.333	[0.002, 0.006]
Chain	0.010^{***}	0.002	0.000	0.392	[0.006, 0.015]
Regulation	0.058^{***}	0.011	0.000	0.437	[0.037, 0.080]
Diagnostic statistics	**Value**				
Observations	234				
χ^2 [p-value]	3.888 [0.143]				
CD	0.608				
CFI	0.989				
TLI	0.960				
RMSEA	0.064				
SRMR	0.022				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

Table 7: Robustness Check: Numbers of Comments mentioned Prepared Meals

Variable	Unstandardized Coefficient	Standard Error	P	Standardized Coefficient	95% CI
Indicators					
N^{PM}	1(constrained)	-	-	0.597	-
Wage Ratio	-0.057^{***}	0.006	0.000	-0.639	$[-0.070, -0.045]$
Factors					
Rent	0.294^{***}	0.040	0.000	0.454	[0.215, 0.373]
Chain	0.738^{***}	0.106	0.000	0.481	[0.530, 0.946]
Regulation	3.397^{***}	0.529	0.000	0.425	[2.360, 4.434]
Diagnostic statistics	**Value**				
Observations	311				
χ^2 [p-value]	1.110 [0.574]				
CD	0.828				
CFI	1.000				
TLI	1.013				
RMSEA	0.000				
SRMR	0.010				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

Table 8: Robustness Check: MIMIC Model Excluding Fast-Food and Hot-Pot Restaurants

Variable	Unstandardized Coefficient	Standard Error	P	Standardized Coefficient	95% CI
Indicators					
Comment Ratio	1(constrained)	-	-	0.652	-
Wage Ratio	-3.588^{***}	0.512	0.000	-0.694	$[-4.592, -2.583]$
Factors					
Rent	0.0039^{***}	0.0008	0.000	0.336	[0.002, 0.006]
Chain	0.0112^{***}	0.0024	0.000	0.405	[0.006, 0.016]
Regulation	0.0611^{***}	0.0117	0.000	0.447	[0.038, 0.084]
Diagnostic statistics	**Value**				
Observations	196				
χ^2 [p-value]	5.939 [0.051]				
CD	0.638				
CFI	0.973				
TLI	0.904				
RMSEA	0.100				
SRMR	0.030				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

Table 9: Placebo Test: Wage Ratio and Consumption Level

Variable		**Coefficient**	**Standard Error**	P	**95% CI**
Wage Ratio		12.314	22.923	0.592	[−32.790, 57.419]
Constant		97.172	31.459	0.002	[35.271, 159.074]
Diagnostic Statistics	Observations	R^2	Adjusted R^2	F-statistics	p-value
Value	311	0.001	−0.002	0.29	0.592

Note: Standard errors are reported in parentheses. The dependent variable is consumption per capita.

Table 10: Robustness Test of External Validation

Group	**Observations**	**Mean**	**Standard Deviation**	**Standard Error**
Known-positive group ($CentralKitchen = 1$)	21	1.022	0.075	0.016
No-public-disclosure group ($CentralKitchen = 0$)	346	1.347	0.311	0.017
Between-group comparison	**Value**			
Mean difference	-0.325			
Welch's t	-13.897			
Degrees of freedom	78.882			
p-value	< 0.001			
95% CI	[−0.371, −0.278]			
Cohen's d	-1.072			
95% CI for Cohen's d	[−1.519, −0.624]			

Note: Mean differences are calculated as $CentralKitchen = 1$ minus $CentralKitchen = 0$. Welch's t-test is reported because Levene's test rejects the equal-variance assumption ($F = 19.457$, $p < 0.001$).